

LECTURE 4 • DAY 2, 10:00–10:30

Introduction to data analysis

SORT IT Module 3 • Data analysis and interpretation of results • Timor-Leste

Analysis is not statistics.

It is deciding what question you are answering — and then counting carefully.

Three rules. That is the whole lecture.

- Know what type of variable you have — before you calculate anything
- Choose the right summary — mean and median are not interchangeable
- State your denominator — every single time, without exception

Rule 1

Know your variable

Get this wrong and every number after it is wrong

The four types

Type	What it is	Example	How you summarise it
Nominal	Categories, no order	Sex, municipality, TB type	n (%)
Ordinal	Categories WITH order	Smear grade, education level	n (%) or median
Discrete	Counted whole numbers	Visits, days, household size	Median (IQR)
Continuous	Measured, can have decimals	Age, weight, distance	Mean (SD) or median (IQR)

What is the average sex?

If your database stores sex as 1 = male and 2 = female, the mean is 1.4.

The Variable Type Race

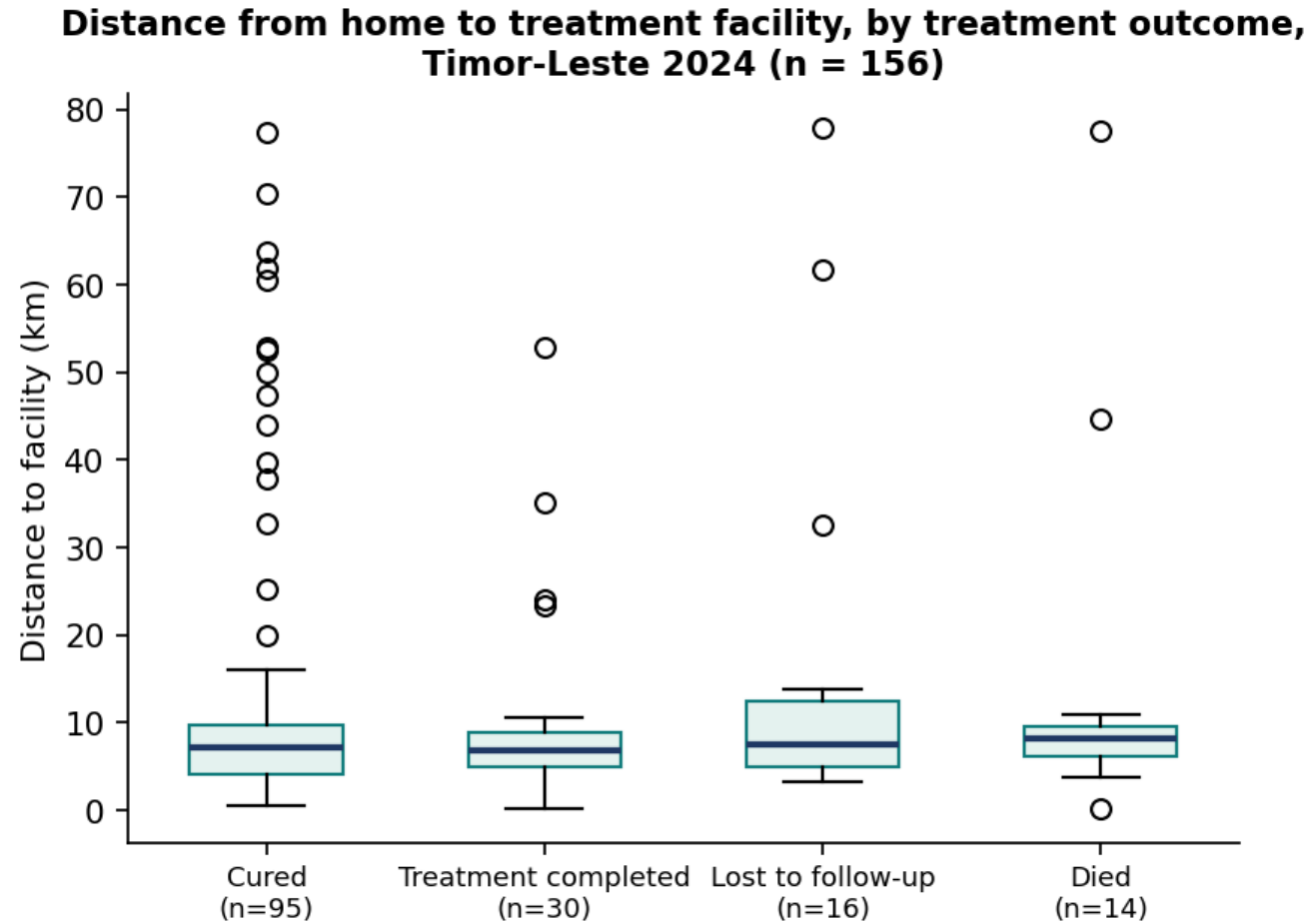
1. In your table, sort all 24 cards under the four headers
2. First table to finish calls out — but do not expect to be right
3. Then challenge your neighbour's sort. Three challenges each
4. 2 bonus points for every successful challenge

Rule 2

Mean or median?

The wrong average tells your reader something untrue

Same 180 patients. Two different 'typical' values.



Distance to facility, by treatment outcome — note how far the boxes overlap

The rule, in one table

Your data are...	Use	Report as
Symmetric	Mean	Mean (SD) — e.g. 48.5 kg (SD 9.1)
Skewed, or has outliers	Median	Median (IQR) — e.g. 7.0 km (4.2–9.8)
Ordinal	Median	Or simply n (%)
Nominal	Neither	n (%) — e.g. 104 (57.8%)

Rule 3

State your denominator

The most common error in operational research — and the most damaging

180 patients. 126 successes. What is the success rate?

Denominator	n	Answer	What you are really saying
All registered	180	70.0%	Includes 8 still on treatment
Excluding still on treatment	172	73.3%	Still includes 12 transferred out
WHO evaluated cohort	160	78.8%	Hides the 12 you lost track of

**A percentage without its denominator
is not a result. It is a rumour.**

Write 126/160 (78.8%). Never 78.8% alone.

The small-denominator trap

- In our data, 13 of 13 HIV-positive patients were treated successfully. That is 100%.
- Would you write that in your paper?
- One more death would make it 92.9%.
- Below $n = 20$ – 30 , report the counts. Skip the percentage entirely.

What happens next

- 11:00 — Group work: the Denominator Hunt, then your own analysis plan
- 12:00 — Lecture 5: using graphs and tables
- 13:30 — Group work: build your tables and charts for real
- Tomorrow 09:00 — recap and interpretation

**Know your variable.
Choose the right average.
State your denominator.**

That is data analysis. Everything else is detail.